# ITA_ØK – Week 8

ITA-ØK. Week 8.
LmStudio. .

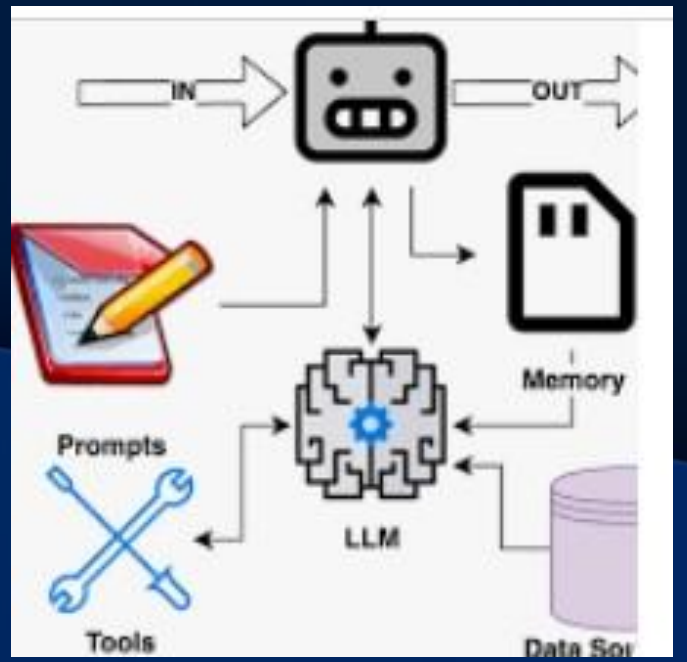


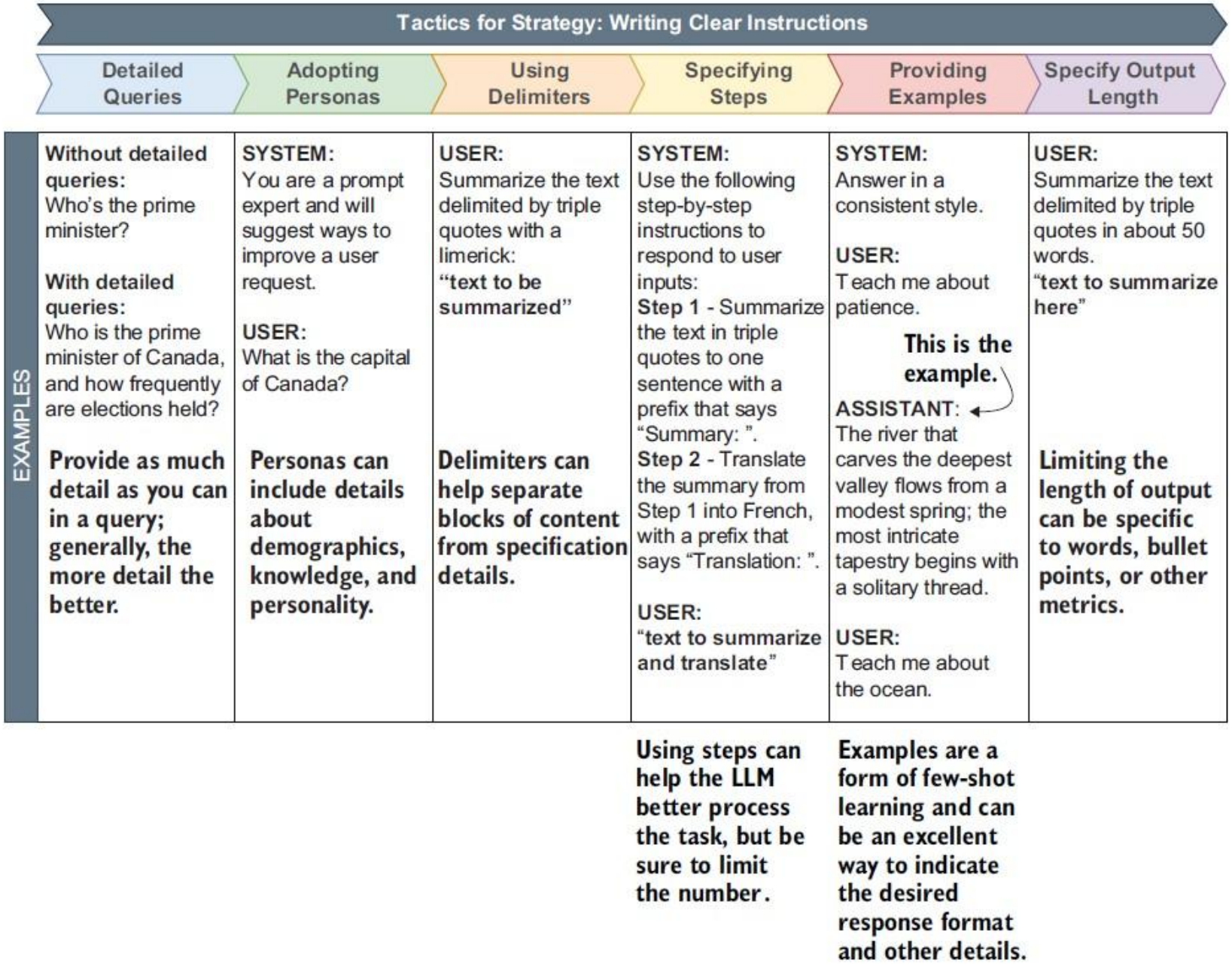
# Today::


AI Agents
python™


Vibe coder


AI Agent
From scratch
Just Python
HANDS-ON DEVELOPMENT


IN
OUT
Memory
Prompts
LLM
Tools




Prompting an LLM.

Clear instructions..

From "AI Agents in Action"
by Michael Lanham.

**Listing 2.9** `detailed_queries.jsonl`

```
[
    {
        "role": "system",
        "content": "You are a helpful assistant."
    },
    {
        "role": "user",
        "content": "What is an agent?"
    }
]
[
    {
        "role": "system",
        "content": "You are a helpful assistant."
    },
    {
        "role": "user",
        "content": """
What is a GPT Agent?
Please give me 3 examples of a GPT agent
"""
    }
]
```

The first example doesn't use detailed queries.

First ask the LLM a very general question.

Ask a more specific question, and ask for examples.

Improve your prompts.

Prompting an LLM.

Clear instructions..

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**Listing 2.10** adopting_personas.jsonl

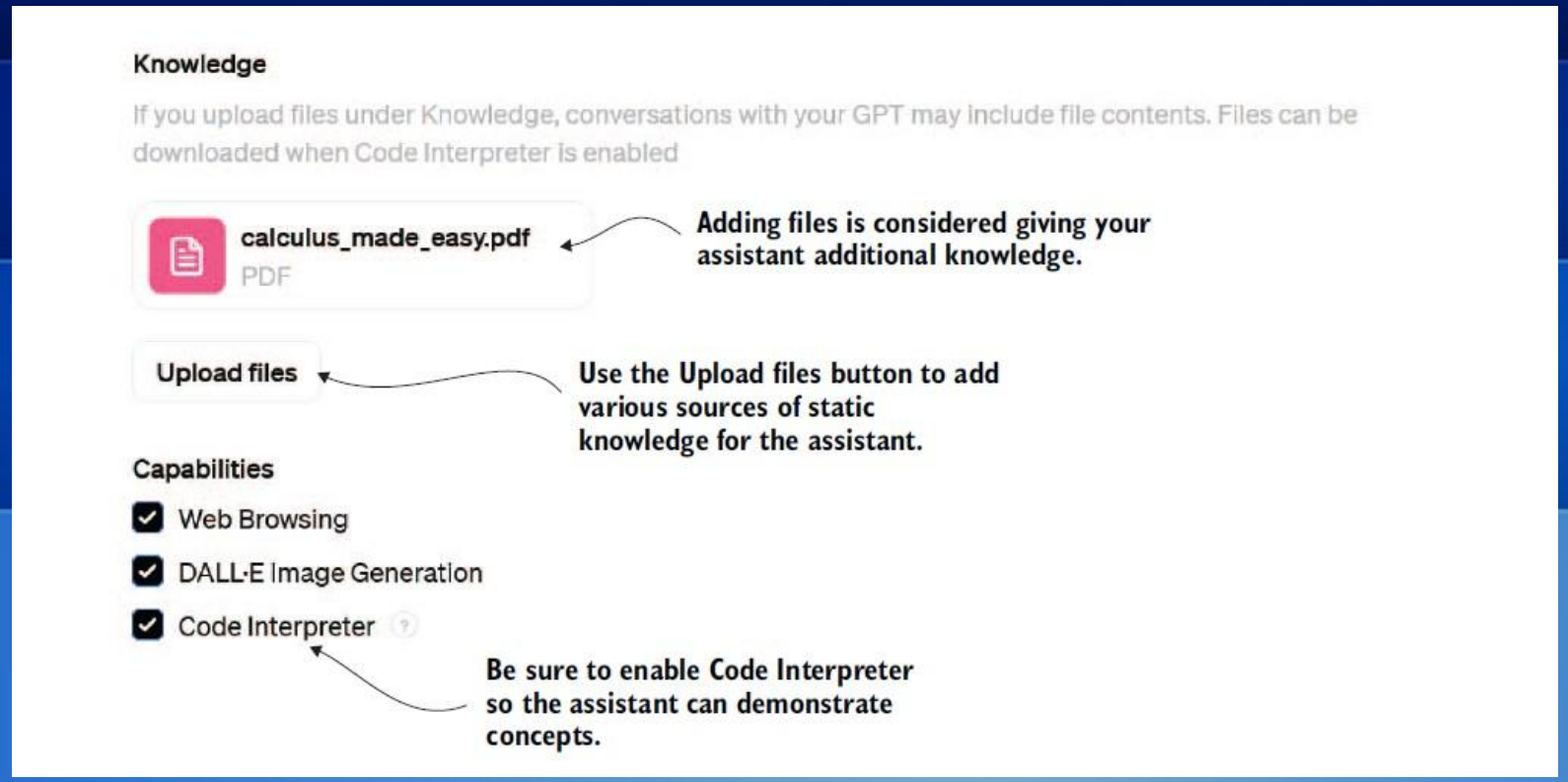
```
[
    {
        "role": "system",
        "content": """
You are a 20 year old female who attends college
in computer science. Answer all your replies as
a junior programmer.
"""                                    <-- First persona
    },
    {
        "role": "user",
        "content": "What is the best subject to study."
    }
]
[
    {
        "role": "system",
        "content": """
You are a 38 year old male registered nurse.
Answer all replies as a medical professional.
"""                                              <-- Second persona
    },
    {
        "role": "user",
        "content": "What is the best subject to study."
    }
]
```

Prompting an LLM.

Clear instructions..

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Improve your prompts.




Prompting an LLM.

Add more knowledge/files to be included in your prompt.

Clear instructions.

**NOTE** If you're serious about learning calculus but this assistant is still too advanced, check out a great book by Clifford A. Pickover called *Calculus and Pizza*. It's a great book for learning calculus or just to get an excellent refresher. You could also try making your Calculus and Pizza assistant if you have an eBook version. Unfortunately, copyright laws would prevent you from publishing this GPT without permission.

Open ChatGPT, go to My GPTs, create a new GPT, click the Configure tab, and then upload the file, as shown in figure 3.10. Upload the book from the chapter's source code folder: `chapter _03/calculus_made_easy.pdf`. This will add the book to the GPT's knowledge.

Scroll up and add the instructions shown in listing 3.16. The initial preamble text was generated by conversing with the GPT Builder. After updating the preamble text, a personality was added by asking ChatGPT for famous mathematicians. Then, finally, rules were added to provide additional guidance to the GPT on what explicit outcomes we want.

Improve your prompts.

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*Objective*—Build an assistant with specialized knowledge from uploaded documents.

*Tasks:*

- Select a freely available e-book or a collection of documents related to a specific topic (e.g., classic literature, technical manuals).
- Upload these files to a new GPT assistant, and configure the assistant to act as an expert on the uploaded content.
- Create a series of prompts to test the assistant's ability to reference and summarize the information from the documents.
- Evaluate the assistant's performance, and make any necessary adjustments to improve its accuracy and helpfulness.

Prompting an LLM.

Add more knowledge/files to be included in your prompt.

Clear instructions.

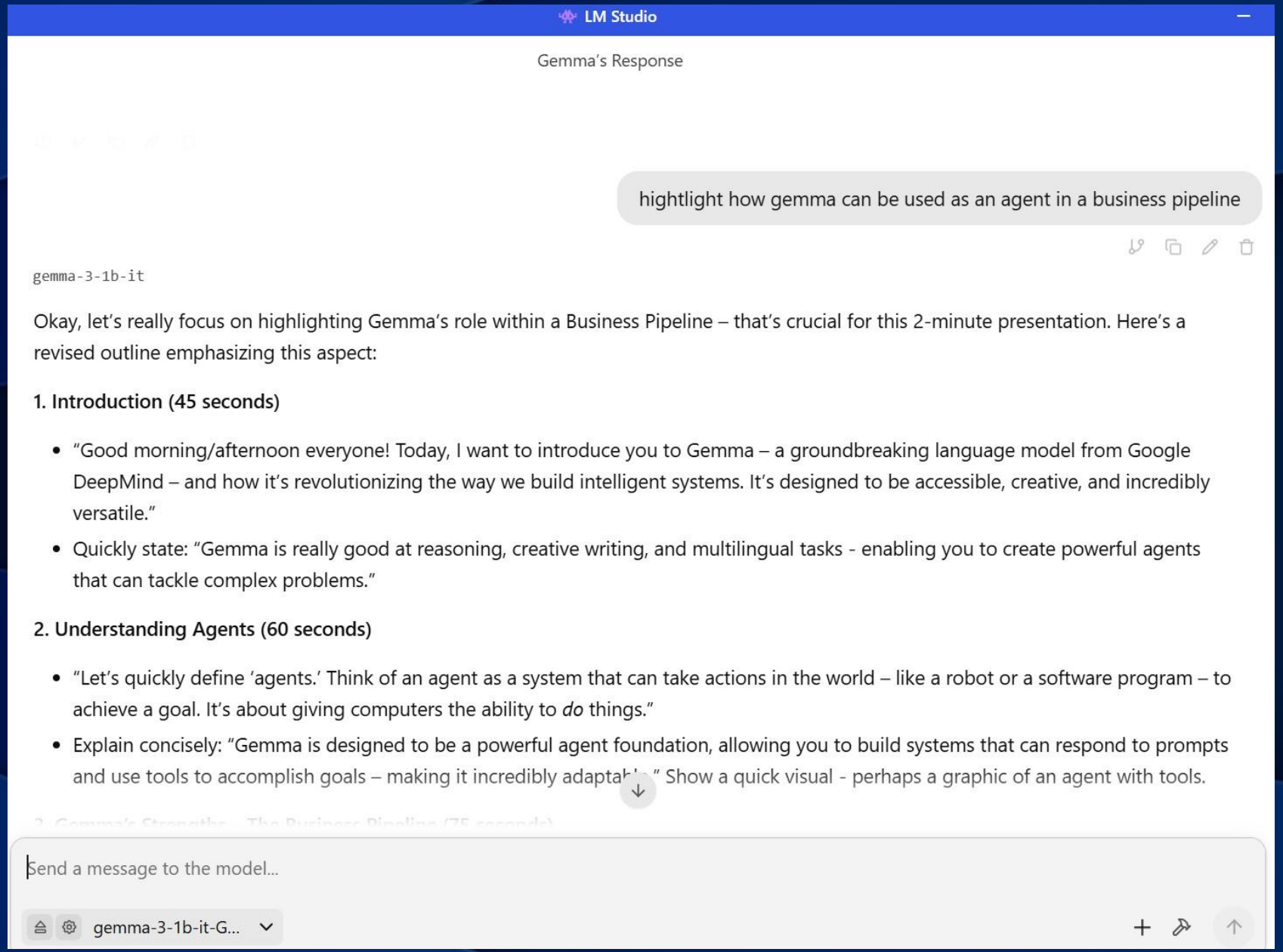
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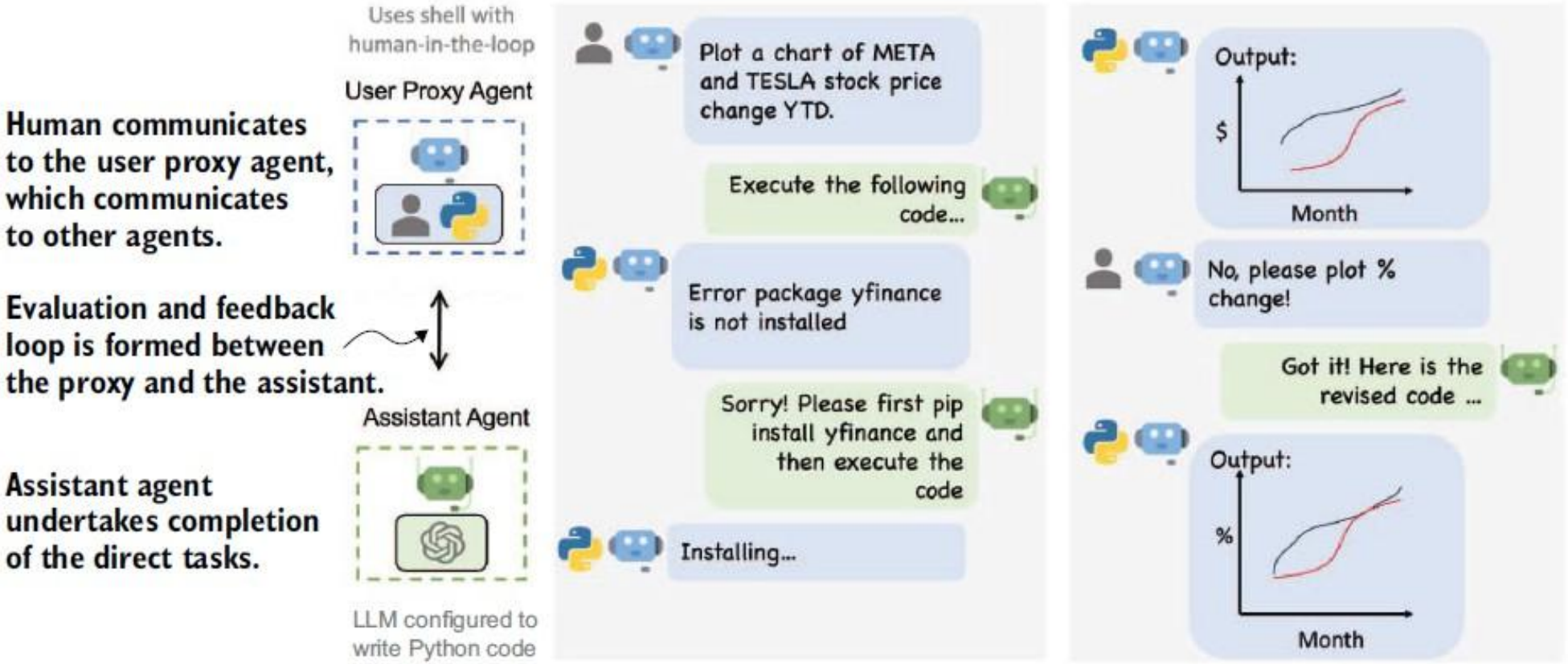
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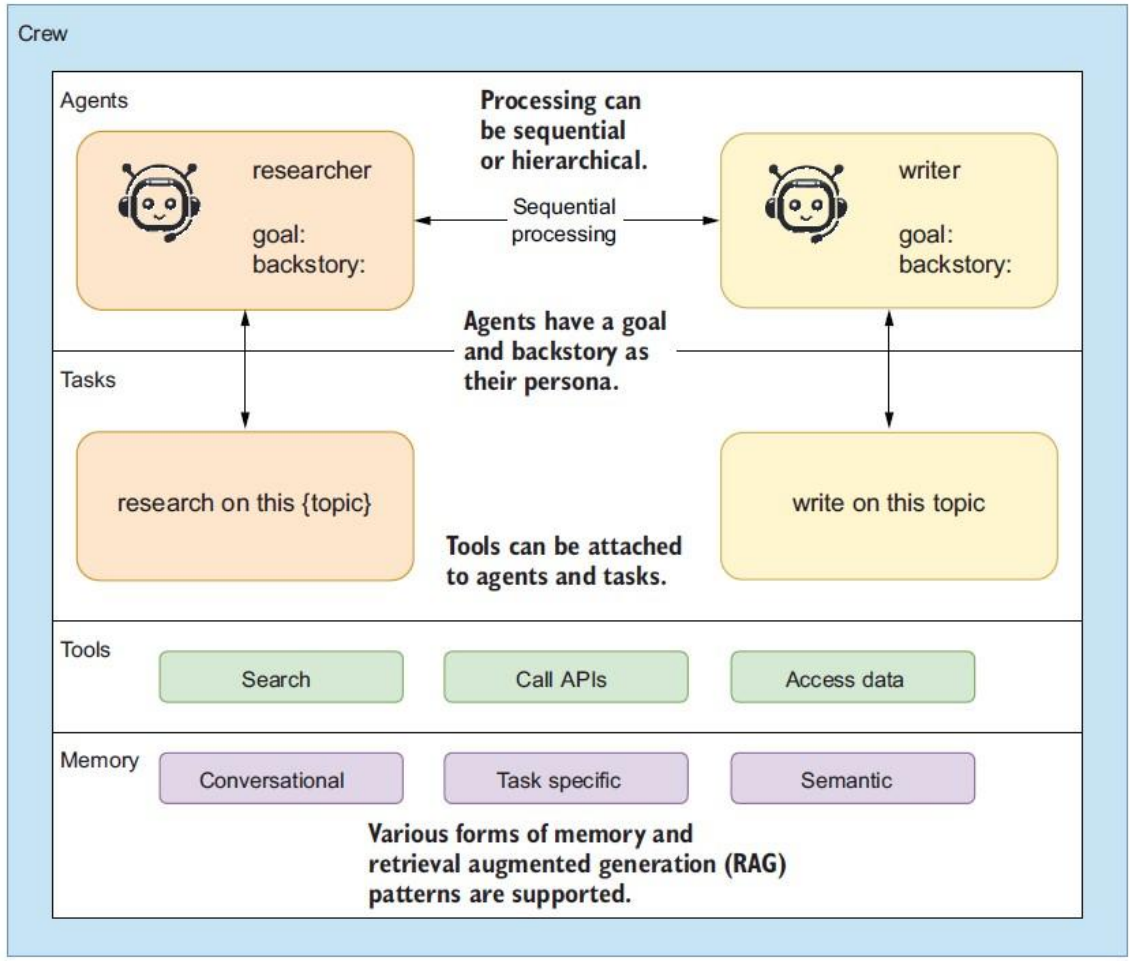

Running LLMs
in LM Studio.

Prompt engineering




More Agents working on a task.

From "AI Agents in Action"
by Michael Lanham.



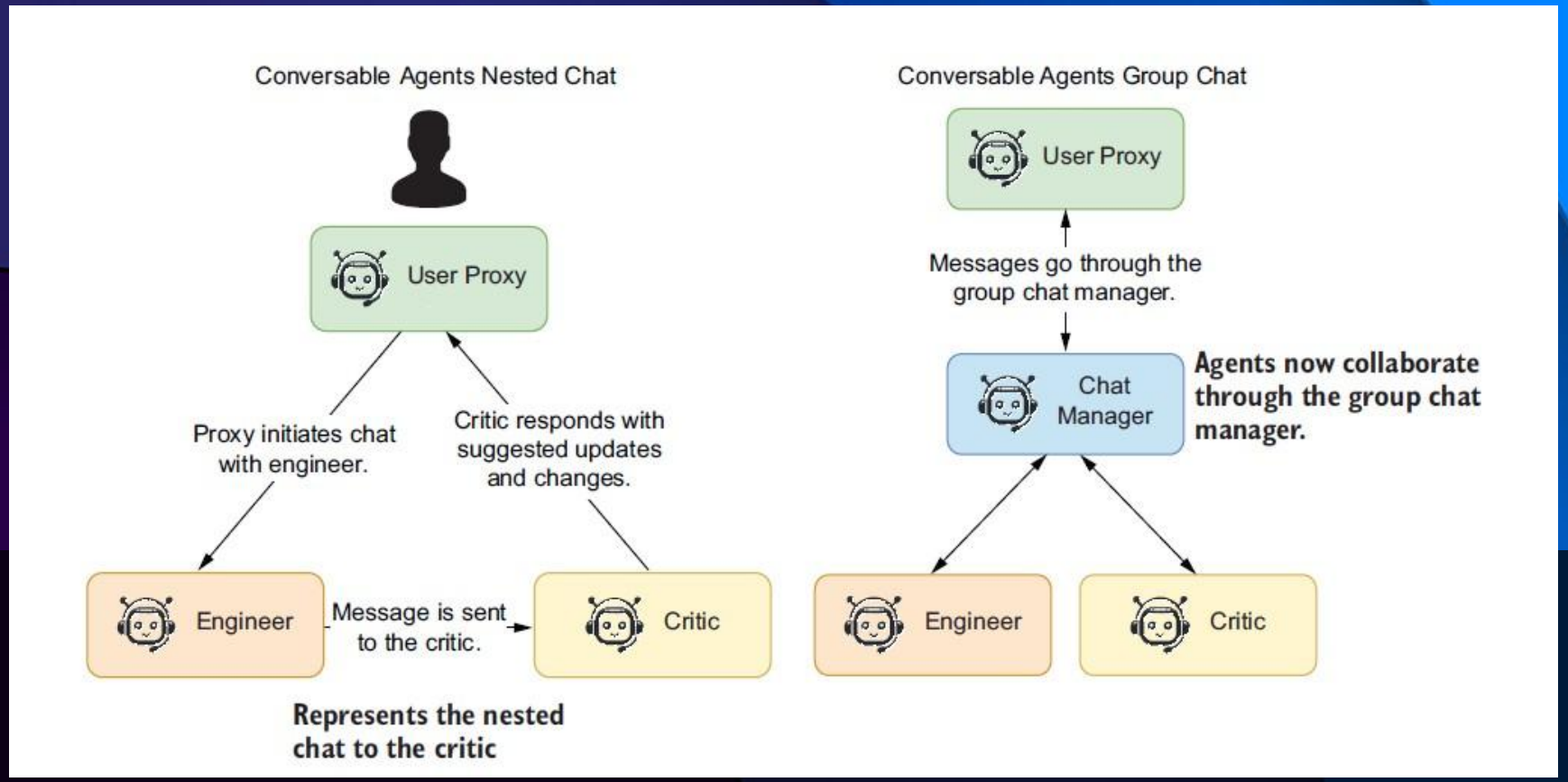



More Agents working on a task.

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***Enhancing code output with agent critics***

One powerful benefit of multi-agent systems is the multiple roles/personas you can automatically assign when completing tasks. Generating or helping to write code can be an excellent advantage to any developer, but what if that code was also reviewed and tested? In the next exercise, we'll add another agent critic to our agent system to help



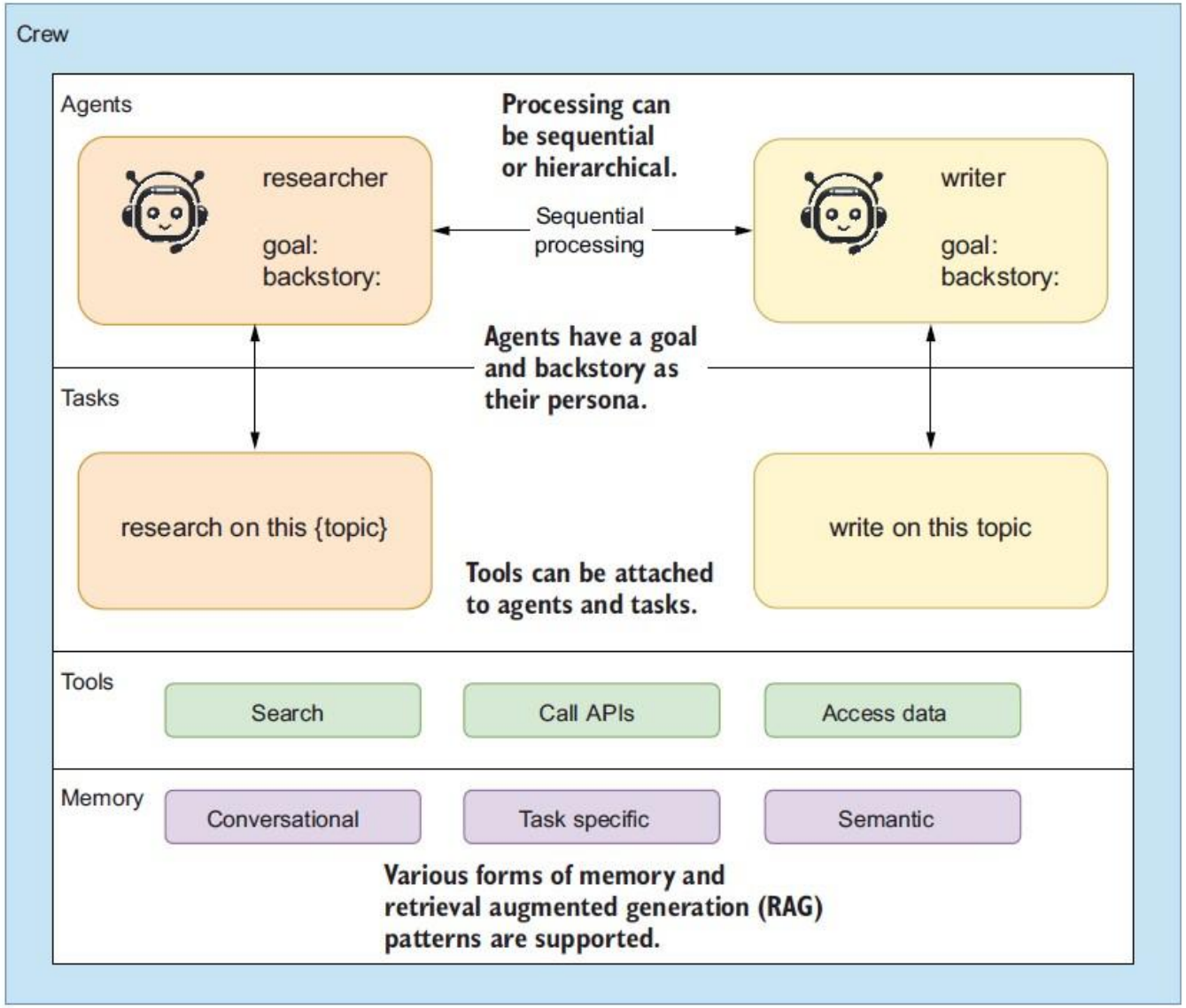



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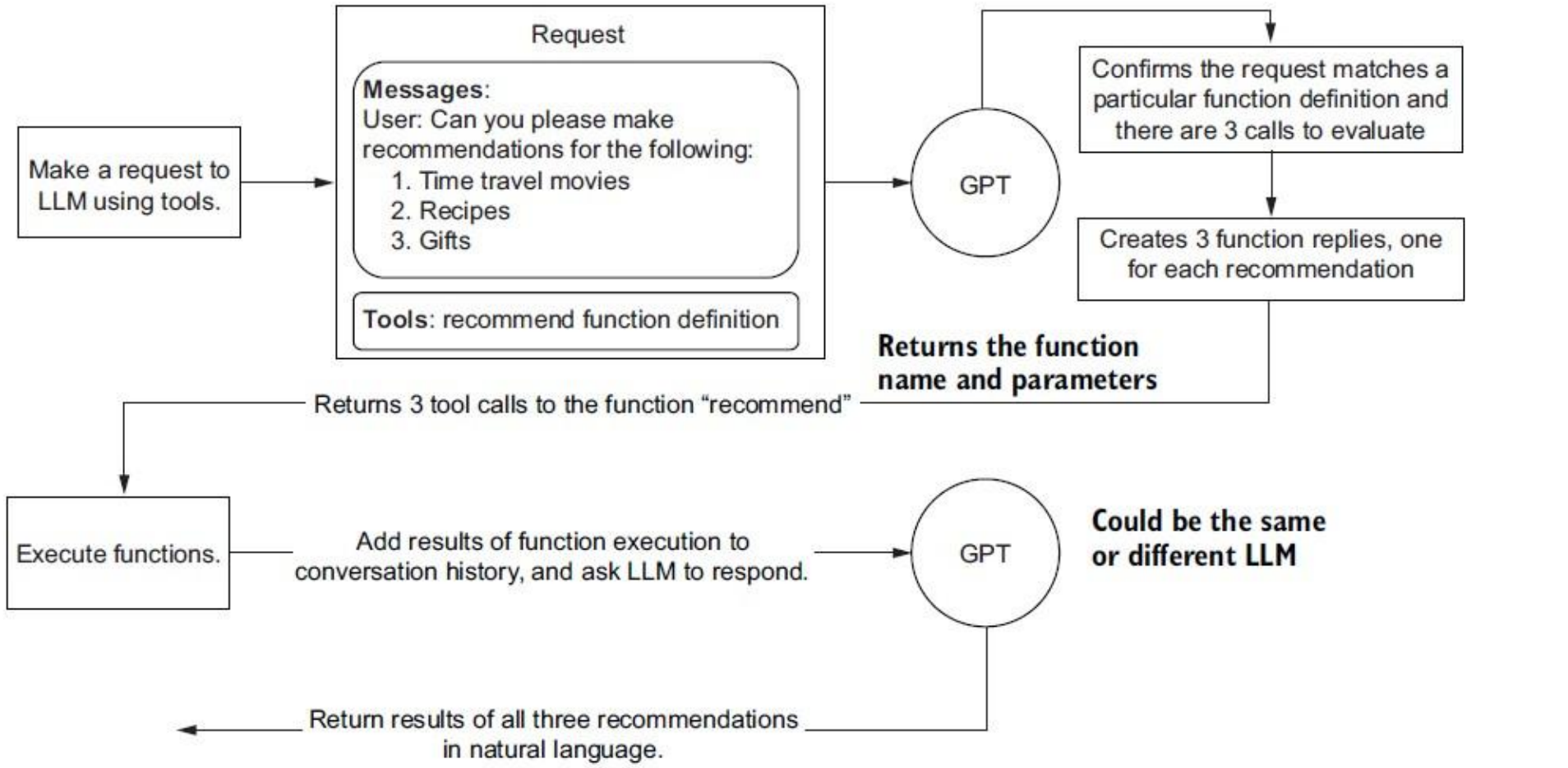
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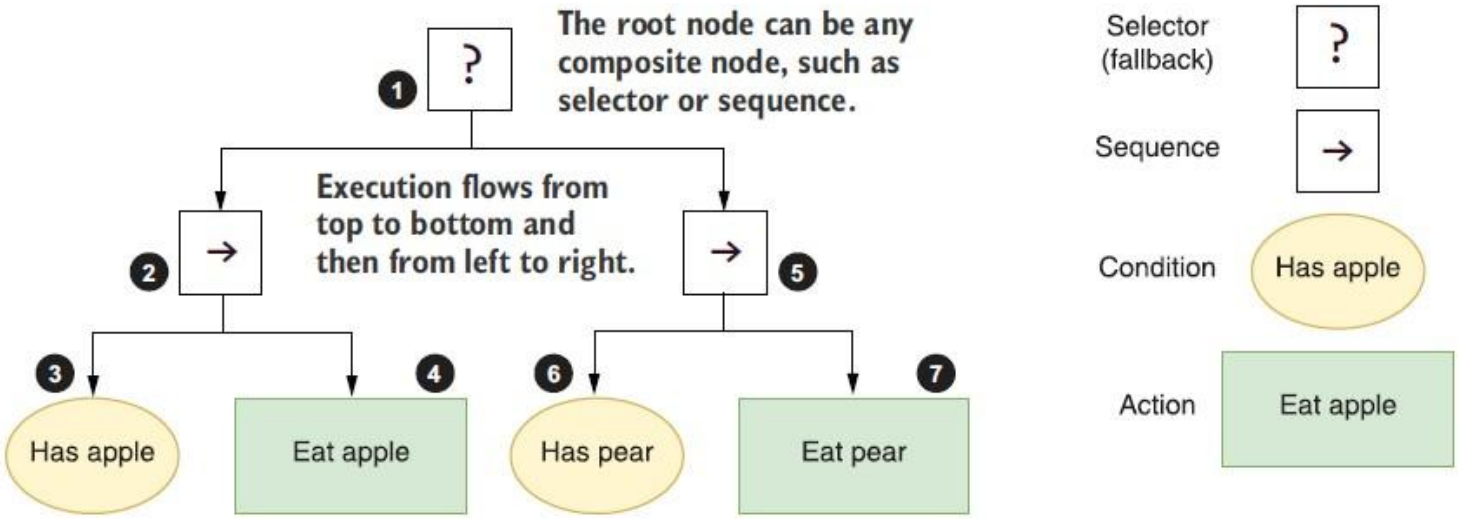



Agents calling functions.

From "AI Agents in Action"
by Michael Lanham.

Behavior trees are a long-established pattern used to control robotics and AI in games. Rodney A. Brooks first introduced the concept in his "A Robust Layered Control System for a Mobile Robot" paper in 1986. This laid the groundwork for a pattern that expanded on using the tree and node structure we have today.

If you've ever played a computer game with nonplayer characters (NPCs) or interacted with advanced robotic systems, you've witnessed behavior trees at work. Figure 6.1 shows a simple behavior tree. The tree represents all the primary nodes: selector or fallback nodes, sequence nodes, action nodes, and condition nodes.




Behaviour Tree calling pattern.

Agents calling functions.

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Understanding how behavior trees execute is crucial to designing and implementing behavior trees. Unlike most concepts in computer science, behavior trees operate in terms of success and failure. When a node in a behavior tree executes, it will return either success or failure; this even applies to conditions and selector nodes.

## *Introducing agentic behavior trees*

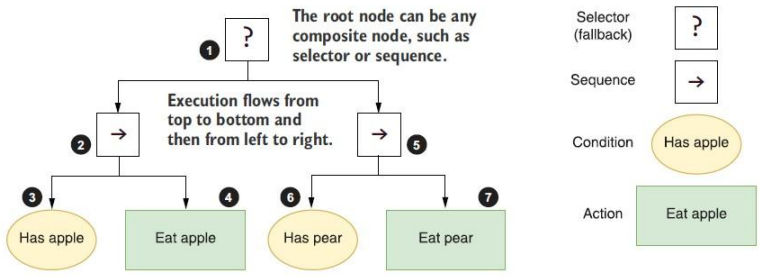
Agentic behavior trees (ABTs) implement behavior trees on assistant and agent systems. The key difference between regular behavior trees and ABTs is that they use prompts to direct actions and conditions. Because prompts may return a high occurrence of random results, we could also name these trees *stochastic* behavior trees, which do exist. For simplicity, we'll differentiate behavior trees used to control agents, referring to them as agentic.

Behaviour Tree calling pattern.

Agents calling functions.

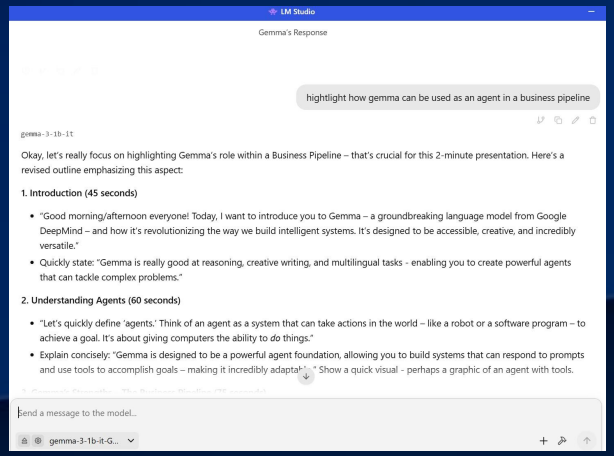
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Running LLMs
In LM Studio.




Prompting LM Studio Server

Prompt engineering

```
C:\>
C:\>curl http://localhost:1234/v1/chat/completions -H "Content-Type: application/json" -d "{\"messages\": [{\"role\": \"
user\", \"content\": \"Say hello!\"}]}"
{
  "id": "chatcmpl-59olfi8272be1guma4gzvb",
  "object": "chat.completion",
  "created": 1773679796,
  "model": "gemma-3-1b-it",
  "choices": [
    {
      "index": 0,
      "message": {
        "role": "assistant",
        "content": "Hello there! How's your day going so far? 😊 \n\nLet me know if you want to chat about anything!",
        "tool_calls": []
      },
      "logprobs": null,
      "finish_reason": "stop"
    }
  ],
  "usage": {
    "prompt_tokens": 12,
    "completion_tokens": 27,
    "total_tokens": 39
  },
  "stats": {},
  "system_fingerprint": "gemma-3-1b-it"
}
C:\>
```

# Exercises
1- 6.